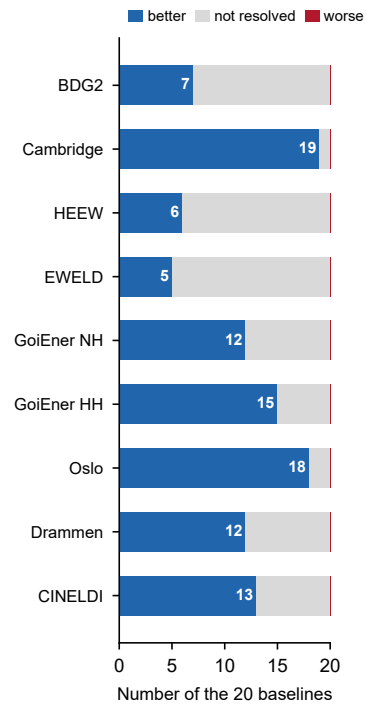
Monthly energy error, 744 × the mean hourly error of the delivered trajectory, on the same test windows (ankyra-lite: its energy readout, the trajectory before the continuation layer, readout rule A). Unit-equal RMS ratio with 95% unit-and-month bootstrap intervals; * resolved in ankyra-lite's favour, (−) against it. BDG2 w/o 3: without the three near-zero meters, where ankyra-lite's error is exactly zero (the point estimates are equal, only the intervals differ). Rows: the 20 baselines (21 forecasters with ankyra-lite), in the order of figure 2. ankyra-lite 0.2.0: carrier recipe Rw (trained from scratch with early stopping on the 93-day validation set, then 8 more epochs with the validation span returned and each training window's loss weighted by the inverse-noise rule; six seeds averaged; figure 1b), the anchoring layer, the state rule and the origin-continuity layer; hourly errors score the delivered trajectory after that layer, energy, level and peak are read before it (readout rule A). Linear-context: the TiDE implementation of the release, whose forecast is a linear map of the load context.

a ankyra-lite's improvement in monthly energy error over each baseline (%)

	BDG2	BDG2 w/o 3	Cambridge	HEEW	EWELD	GoiEner NH	GoiEner HH	Oslo	Drammen	CINELDI
Ridge (per unit)	+11	+11	**+23***	+2	+5	+8	+2	**+33***	-5	+10
Chronos-2-X	+3	+3	**+20***	-1	+17	+12	**+33***	**+20***	+4	+17
iTransformer-X	**+21***	**+21***	+8	-1	+5	+17	+16	+10	+9	+13
iTransformer	+19	+19	**+20***	+2	+2	+10	**+22***	**+25***	+9	**+15***
PatchTST	+15	+15	**+20***	+5	-5	+14	**+18***	**+24***	+9	+9
GBT-T	+11	+11	**+23***	+3	+25	**+44***	**+21***	+6	**+14***	**+25***
TimesFM-X	+18	+18	**+26***	+4	-2	**+18***	**+29***	**+39***	**+23***	**+20***
TimesFM	+15	+15	**+25***	0	-3	**+20***	**+33***	**+35***	**+29***	**+22***
Chronos-2	+15	+15	**+31***	+6	-2	+18	**+34***	**+37***	**+27***	+20
DLinear	+10	+10	**+39***	**+27***	**+27***	**+25***	**+19***	**+42***	+12	**+16***
Linear-context	+9	+9	**+41***	**+27***	**+26***	**+26***	**+21***	**+42***	+12	**+16***
4-week profile	**+31***	**+31***	**+25***	+21	+5	**+32***	+3	**+43***	**+44***	+15
Holt–Winters	**+39***	**+39***	**+39***	+20	**+37***	**+39***	**+39***	**+50***	**+38***	**+41***
LSTM	**+19***	**+19***	**+47***	**+34***	**+37***	**+20***	**+17***	**+54***	+23	**+28***
Previous-month profile	+4	+4	**+27***	**+23***	+5	**+35***	+1	**+47***	**+44***	+17
MSTL	+26	+26	**+40***	+5	+20	**+34***	**+40***	**+51***	**+36***	**+43***
Seasonal naive (week)	+21	+21	**+32***	+7	0	+16	+7	**+45***	**+30***	**+24***
Last-year profile	**+19***	**+19***	**+55***	**+49***	**+58***	+21	**+24***	**+32***	**+35***	**+37***
GBT (zero-shot)	**+39***	**+39***	**+29***	+19	+13	**+42***	**+44***	**+49***	**+38***	**+26***
Seasonal naive (day)	**+45***	**+45***	**+52***	**+22***	+19	**+31***	**+34***	**+61***	**+54***	**+56***

b Head-to-head on energy



Same information · trained
Same information · zero-shot foundation model
Load only · zero-shot foundation model
Load only · statistical
Load only · trained
Per-unit regression (calendar, temperature)
Profiles, naive, zero-shot GBT